

SECTION J - DIFFERENTIATION

Differentiate the following functions.

1. $f'(x) = [\cosh(x^2 + 4x)] \cdot 2x$
2. $f'(x) = [\cosh(7x)] \cdot 7$
3. $f'(x) = [\sinh(x^4 + 5x + 7)] \cdot [4x^3 + 5]$
4. $f'(t) = [\operatorname{sech}^2(\sinh 2t)] \cdot [\cosh(2t)] \cdot 2$
5. $f'(x) = [\operatorname{sech}^2(x^2 + 5)] \cdot 2x$
6. $f'(x) = \frac{[\cosh x]x - \sinh x}{x^2}$
7. $f'(y) = [\operatorname{sech}^2(2y - y^3)] \cdot (2 - 3y^2)$
8. $f'(x) = 3[\cosh(x^2)] \cdot 2x + 2 \sinh(x)$
9. $f'(x) = 3[\cosh 4x] \cdot 4 - 2[\sinh 5x] \cdot 5$
10. $f'(x) = [\sinh(\tanh x)] \cdot \operatorname{sech}^2 x$
11. $f'(x) = \left[\sinh \left(\frac{1}{\sqrt{x}} \right) \right] \cdot \left(-\frac{1}{2} \right) x^{-\frac{3}{2}}$
12. $f'(x) = \left[\operatorname{sech}^2 \left(\frac{1}{x} \right) \right] \cdot \frac{-1}{x^2}$
13. $f'(t) = -\cosh(-t)$
14. $f'(x) = \frac{[\sinh x](1 + x) - \cosh x}{(1 + x)^2}$
15. $f'(x) = 4 \cdot 2[\cosh(3x)](\sinh(3x)) \cdot 3$
16. $f'(x) = 2[\sinh(x^4)] [\cosh(x^4)] \cdot 4x^3$
17. $\frac{d}{dx} \ln e^{\cosh 2x} = \frac{d}{dx} \cosh 2x = 2 \sinh 2x$
18. $f'(\theta) = \frac{2}{\sqrt{1 - (2\theta)^2}}$
19. $f'(x) = \frac{-2x}{\sqrt{1 - (x^2)^2}}$
20. $f'(x) = 1 \times (\arcsin x) + x \times \frac{1}{\sqrt{1 - x^2}}$
21. $f'(x) = \frac{\left(\frac{-2}{x^2} \right)}{\sqrt{1 - \left(\frac{2}{x} \right)^2}}$
22. $f'(y) = \frac{2}{\sqrt{1 - (2y - 3)^2}}$
23. $f'(x) = 2(\arctan x) + 2x \cdot \frac{1}{1 + x^2}$

$$24. f'(x) = \frac{5}{1 + (5x)^2}$$

$$25. f'(t) = \frac{\cos t}{\sqrt{1 - (\sin t)^2}} = 1$$

$$26. f'(x) = \frac{3}{1 + (3x - 4)^2}$$

$$27. f'(x) = \frac{-\frac{1}{4}}{\sqrt{1 - \left(\frac{x}{4}\right)^2}}$$